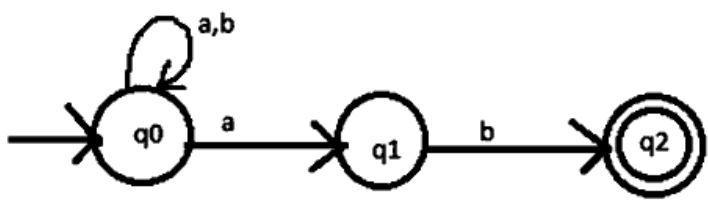


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination – 2023 Course: B. Tech. Branch :Computer Engineering Semester :V Subject Code & Name: Theory Of Computation (BTCOC502) Max Marks: 60 Date:09/08/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Construct a Deterministic Finite Automaton for the set of string over {a, b} such that length of the string $ w =2$.	Design	6
B)	State and explain applications of Regular Expressions.	Understand	6
C)	Design an NFA with $\Sigma = \{0, 1\}$ accepts all string ending with 01.	Design	6
Q.2	Solve Any Two of the following.		12
A)	Distinguish between Moore machine and Mealy machine .	Analyze	6
B)	Write a regular expression over alphabet $\Sigma = \{0, 1\}$ for following . (i) begins with 1, ends with 1 (ii) ends with 00 (iii) contains at least three consecutive 1s	Create	6
C)	Convert the following Non Deterministic Automaton into Deterministic Finite Automaton . 	Apply	6
Q. 3	Solve Any Two of the following.		12
A)	Write the steps for grammar reduction. Eliminate Unit Productions from the following grammar. $S \rightarrow Aa B c$ $B \rightarrow A bb$ $A \rightarrow a bc B$	Apply	6

B)	Explain Chomsky classification of grammar.	Remember	6
C)	When does context free grammar is said to be in Chomsky Normal Form(CNF)? Write steps to convert context free grammar into CNF.	Understand	6
Q.4	Solve Any Two of the following.		12
A)	Write the productions rule of Context free grammar for following regular expressions. (i) 0^* (ii) $(a+b)^*$ (iii) $(ab)^*$	Design	6
B)	What are the different components of Pushdown Automaton? Explain with neat diagram.	Remember	6
C)	Distinguish between Deterministic and Non Deterministic PDA.	Analyze	6
Q. 5	Solve Any Two of the following.		12
A)	Design Turing machine that accepts the language of all strings over alphabet $\Sigma = \{a, b\}$ whose second letter is b.	Design	6
B)	Explain the following (i)Multihead Turing machine (ii)Universal Turing machine (iii) Non Deterministic Turing machine	Understand	6
C)	What is Church Turing Thesis ? Explain.	Remember	6
	*** End ***		

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